PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. under the Panjab University Act VII of 1947 – enacted by the Govt. of India)

SYLLABI

COURSE OUTLINE FOR
B.VOC (FOOD PROCESSING & QUALITY MANAGEMENT)
AS PER NEP 2020

W.E.F SESSION 2026-27

COURSE OUTLINE FOR B.VOC
(FOOD PROCESSING & QUALITY MANAGEMENT)
AS PER NEP 2020 W.E.F. SESSION 2025-26

SEMESTER-I										
Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
FPQM 101	Core Discipline	Food Biochemistry	2	2	4	40	10	40	10	100
FPQM 102	Core Discipline	Dairy Technology	2	2	4	40	10	40	10	100
FPQM 103	Core Discipline	Food Quality Control	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English	-	-	2	-	-	-	-	50
SEC I	Skill Enhancement Course	Common Basket	-	-	3	-	-	-	-	75
IDC I	Interdisciplinary Course	Common Basket	-	-	3	-	-	-	-	75
VAC I	Value-Added Course	Common Basket	-	-	2	-	-	-	-	50
PW I	Project Work	Project Work-I & Viva-Voce	-	2	2	-	-	50	-	50
		Total Credits			24					600

SEMESTER-II										
Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
FPQM 201	Core Discipline	Food Packaging-1	2	2	4	40	10	40	10	100
FPQM 202	Core Discipline	Principles of Food Processing and Preservation	2	2	4	40	10	40	10	100
FPQM 203	Core Discipline	Food Microbiology	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English	-	-	2	-	-	-	-	50
SEC II	Skill Enhancement Course	Common Basket	-	-	3	-	-	-	-	75
IDC II	Interdisciplinary Course	Common Basket	-	-	3	-	-	-	-	75
SIT 201*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
		Total Credits			24					600

Job Role :- Dairy and Food Processor.

* Summer Industrial Training of 2-4 weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 3rd Semester. Viva-Voce examination to be held within 3-weeks of the start of 3rd semester.

** Minimum Credits required to get Certificate Course in B.Voc. Food Processing & Quality Management

48 + 4 #.

#Exit 1: Certificate Course - Student can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.

Semester-III										
Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
FPQM 301	Core Discipline	Post-Harvest Management of Fruits & Vegetables	2	2	4	40	10	40	10	100
FPQM 302	Core Discipline	Egg, poultry and fish processing	2	2	4	40	10	40	10	100
FPQM 303	Core Discipline	Cereal Milling & Legumes	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English	-	-	2	-	-	-	-	50
SEC III	Skill Enhancement Course	Common Basket	-	-	3	-	-	-	-	75
IDC III	Interdisciplinary Course	Common Basket	-	-	3	-	-	-	-	75
VAC II	Value-Added Course	Common Basket	-	-	2	-	-	-	-	50
PW II	Project Work	Project Work-II & Viva-Voce	-	2	2	-	-	50	-	50
		Total Credits			24					600

Semester-IV										
Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
FPQM 401	Core Discipline	Technology of oils and fats	2	2	4	40	10	40	10	100
FPQM 402	Core Discipline	Fruit and Vegetable Processing	2	2	4	40	10	40	10	100
FPQM 403	Core Discipline	Processing of Meat & Meat Products	2	2	4	40	10	40	10	100
FPQM 404	Core Discipline	Processing of Cereals	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English/MIL	-	-	2	-	-	-	-	50
VAC III	Value-Added Course	Common Basket	-	-	2	-	-	-	-	50
SIT 401	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
		Total Credits			24					600

Job Role: Egg, Meat & Cereal Specialist.

* Summer Industrial Training of 2-4 weeks in a relevant Industry after 4th Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 3rd Semester. Viva-Voce examination to be held within 3-weeks of the start of 3rd semester.

** Minimum Credits Required to get Diploma Course in B.Voc. Food Processing & Quality Management
= 96 + 4 #.

#Exit 2: Diploma Course - Student can exit after completion of 2nd year having secured 96 Credits and will be awarded a UG Diploma provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 9 credits from skill enhancement courses earned during first, second and third semester.

SEMESTER-V										
Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
FPQM 501	Core Discipline	Technology of fermented Foods – I	2	2	4	40	10	40	10	100
FPQM 502	Core Discipline	Food Analysis	2	2	4	40	10	40	10	100
FPQM 503	Core Discipline	Nutraceuticals & Functional Foods	2	2	4	40	10	40	10	100
FPQM 504^	Core Discipline	Technology of Spices & flavors	2	2	4*	40	10	40	10	100
FPQM 505^	Core Discipline	Unit Operations in Food Engineering	2	2	4*	40	10	40	10	100
FPQM 506^	Core Discipline	Food Safety and Food Laws	2	2	4*	40	10	40	10	100
VAC IV	Value-Added Course	Common Basket			2					50
PW III	Project Work	Project Work – III Viva-Voce			2					50
		Total Credits			24					600

^Any two in 5th Semester

SEMESTER-VI										
Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
FPQM 601	Core Discipline	Food Plant layout	2	2	4	40	10	40	10	100
FPQM 602	Core Discipline	Food Additives	2	2	4	40	10	40	10	100
FPQM 603	Core Discipline	Fundamentals of Food Nutrition	2	2	4	40	10	40	10	100
FPQM 604	Core Discipline	Food Beverage Technology	2	2	4	40	10	40	10	100
FPQM 605^	Core Discipline	Confectionary & Sugar Technology	2	2	4*	40	10	40	10	100*
FPQM 606^	Core Discipline	Grain Storage	2	2	4*	40	10	40	10	100*
FPQM 607^	Core Discipline	Enzyme in Food Industry	2	2	4*	40	10	40	10	100*
		Total Credits			24					600

^Any two in 6th Semester

Job Role : Food Analyst

SEMESTER-VII										
Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
FPQM 701	Core Discipline	Food Packaging -2	2	2	4	40	10	40	10	100
FPQM 702	Core Discipline	Traditional Indian Dairy Products	2	2	4	40	10	40	10	100
FPQM 703	Core Discipline	Food Biochemistry-2	2	2	4	40	10	40	10	100
FPQM 704^	Core Discipline	Unit operations in food engineering -2	2	2	4*	40	10	40	10	100*
FPQM 705^	Core Discipline	Food Plant Hygiene & sanitation	2	2	4*	40	10	40	10	100*
FPQM 706^	Core Discipline	Quality Assurance	2	2	4*	40	10	40	10	100*
SIT 701	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
		Total Credits			24					600

^Any two in 5th Semester

SEMESTER-VIII										
Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
FPQM 801	Core Discipline	Technology of Fermented Foods -2	2	2	4	40	10	40	10	100
FPQM 802	Core Discipline	Industrial Safety & Hazards	2	2	4	40	10	40	10	100
FPQM 803	Core Discipline	Malting & Brewing Technology	2	2	4	40	10	40	10	100
FPQM 804	Core Discipline	Waste Management & by Product Utilization	2	2	4	40	10	40	10	100*
FPQM 805^	Core Discipline	Food Processing Instruments	2	2	4*	40	10	40	10	100*
FPQM 806^	Core Discipline	Industrial Microbiology	2	2	4*	40	10	40	10	100*
FPQM 807^	Core Discipline	Technology of oils & fats-2	2	2	4*	40	10	40	10	100
		Total Credits			24					600

^Any two in 8th Semester

Job Role : Food safety Executive

PROGRAMME OUTCOMES

PO1. Students know about the scientific method to food science problems.

PO2. Know about quantitative reasoning skills to food science data.

PO3. Apply critical thinking and analytical evaluation to contemporary food science information and literature.

PO4. Apply principles from general chemistry, biology, physics, stastics and mathematics to food science problems.

SEMESTER I

SEMESTER I

<table>
<tr><td>1</td><td>Course Title</td><td>Introduction to Biochemistry</td></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM-101</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration / LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 6 0 hours of practical.
(1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
100 Marks (80 TH / PR + 20 IA)
a) The exam of this course will be of 3 hours duration.
b) There will be no objective type questions.
c) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td></td><td>Job Role</td><td>Dairy & Food Processor</td></tr>
<tr><td>7</td><td>Course Objective</td><td>1. Study of various nutrients and maintaining the biochemical properties of food products.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
<table>
<tr><td>CO1</td><td>Understand the role of different nutrients in various food products.</td></tr>
<tr><td>CO2</td><td>Know the sources, classification and properties of various nutrients in food products.</td></tr>
<tr><td>CO3</td><td>Study the structure of various types of nutrients.</td></tr>
</table>
</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.</td></tr>
</table>

		3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.	
11	**Course Content**	UNIT-I	General introduction and importance of Biochemistry. Different nutrients of food, Carbohydrates - introduction, sources, classification, basic structure, biochemical functions and metabolic pathways of carbohydrates {Glycolysis and Kreb cycle)
		UNIT-II	Proteins - Introduction, classification, sources, structural organisations and biochemical functions. Amino acids and their importance (Essential and Non- Essential Amino acids) and properties of amino acids.
		UNIT-III	Lipids - Introduction, classification, sources and biochemical functions. Brief introduction to fatty acids, essential fatty acids, their physical and chemical properties. Oxidation and Rancidity.
		UNIT-IV	Vitamins - Introduction, classification, sources, biochemical functions, daily requirements & deficiency diseases of fat and water soluble vitamins. Minerals - Introduction, classification as major and minor elements, sources, biochemical functions, daily requirements & deficiency diseases of Ca, Fe, I, P, Na, K, F & Zn.
		PRACTICAL WORK:	
		1. Preparation of standard solutions of acid and alkali. 2. Determination of pH using indicators and with pH meter. 3. Determination of moisture in a given sample. 4. Determination of ash in a given sample. 5. Determination of acidity in a given sample. 6. Estimation of fat by soxhlet method. 7. Estimation of free fatty acid of given sample.	
12	**Suggested Resources for Reading**	1. Principles of Biochemistry by Lehinger AL, 1995, CBS Publisher, New Delhi. 2. Text Book of Biochemistry by Rama Rao AVSS, 1999, LK and S Publishers, Visakhapatanm, Indian. 3. Laboratory Manual in Biochemistry by Jayaraman J, 1996, New Age International Ltd., New Delhi	
NOTE LATEST EDITION OF TEXT BOOKS MAY BE USED			

<table>
<tr><td>1</td><td>Course Title</td><td>Dairy Technology</td></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM-102</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration / LP
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d) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td></td><td>Job Role</td><td>Dairy & Food Process</td></tr>
<tr><td>7</td><td>Course Objective</td><td>1. To supervise manufacture of dairy products like milk, butter, yogurt, cheese and their preservation.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
<table>
<tr><td>CO1</td><td>To develop knowledge among students about various aspects of dairy industry.</td></tr>
<tr><td>CO2</td><td>To study quality standards and production of various types of milk and milk products.</td></tr>
<tr><td>CO3</td><td>To study the role of dairy farming in Indian economy</td></tr>
<tr><td>CO4</td><td></td></tr>
</table>
</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
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		3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.	
11	**Course Content**	UNIT-I	Milk: Definition, composition, physicochemical properties of milk and its constituents, nutritional value, overview of processes used in milk and milk products- pasteurization, homogenization and standardization, dye detection test, platform tests, milk borne diseases. Microbiology of milk, types of spoilage and its preventions. Types of market milk: Toned milk, Double Toned milk, Standardized Milk, Recombined milk, reconstituted milk, fortified milk. Effluent treatment of dairy plant disposal.
		UNIT-II	Technology underlying manufacture of Milk products: Butter, cream, butteroil, cheese, yoghurt, paneer, chhana, ice-cream
		UNIT-III	Composition, standards, manufacturing: Process, equipments and defects during manufacturing and storage of dairy products and by products (paneer, dahi, milk powder-skimmed milk and whole milk powder, casein, whey concentrate, lactose, ghee residue).
		UNIT-IV	Dairy development in India: - Present status, future prospective and its role in Indian economy. Important government initiative. Role of dairy development organizations (NDRI, Amul) in dairy development
		PRACTICAL WORK:	
		1. Sampling of milk and milk products for microbiological analysis 2. Platform test for milk analysis. 3. To determine fat in milk by gerber method. 4. To determine specific gravity of given sample of milk by lactometer. 5. Visit and study a nearby milk union/ dairy and prepare a checklist of problems in procurement and milk distribution. 6. Detection of various adulterants in milk. 7. Preparation of sterilized flavored milk. 8. To prepare pasteurized milk. 9. Methylene blue reduction test in milk	
12	**Suggested Resources for**	1. Fluid milk industry, J.S Handerson, A.V.I Publishing Company, USA1	

	Reading	2. Indian Dairy products, K.T. AcharyaPublication 3. Milk Hygiene in milk production processing and distribution, F.A.O Publication. 4. Sukumar.1983.Outlines of Dairy Technology. De,Oxford University Press.
NOTE LATEST EDITION OF TEXT BOOKS MAY BE USED		

<table>
<tr><td>1</td><td>Course Title</td><td>Food Quality Control</td></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM-103</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration / LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 6 0 hours of practical.
(1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
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100 Marks (80 TH / PR + 20 IA)
a) The exam of this course will be of 3 hours duration.
b) There will be no objective type questions.
c) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
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<tr><td></td><td>Job Role</td><td>Dairy & Food Processor</td></tr>
<tr><td>7</td><td>Course Objective</td><td>1. To determine and establish quality standards for food products and Is responsible for ensuring that food products produced meet standards set by both the organization and regulatory authorities.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
<table>
<tr><td>CO1</td><td>To understand the different principles and functions of food quality control department.</td></tr>
<tr><td>CO2</td><td>To understand various food laws and regulations</td></tr>
<tr><td>CO3</td><td>Understand the role of food quality assurance</td></tr>
<tr><td>CO4</td><td>Understand the principles of total quality management systems & standards in India.</td></tr>
<tr><td>CO5</td><td>Understand the principles of total quality management.</td></tr>
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<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
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		comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.	
11	**Course Content**	UNIT-I	Objectives, functions and principles of quality control. Difference between food quality control and quality assurance, assessment of raw materials and finished products (quality control)of baking industry.
		UNIT-II	Total quality management (TQM) – Principles of quality management, good manufacturing practices, good hygienic practices, good lab practices, general awareness and role of management practices in quality control. Quality Attributes: Size, shape, colour, aroma and texture. Microbial quality control: an overview
		UNIT-III	Food laws and regulations, grade and standards, concepts of Codex Almentarious, HACCP, USFDA, ISO 9000 series etc, Food standards and safety Act: salient provisions and prospects, role of various food standards in India- PFA, FPO, AGMARK and BIS.
		UNIT-IV	Sensory quality evaluation: Introduction, methods, panel screening, selection methods, Sensory and instrumental analysis in quality control. Food adulteration, nature of adulteration, methods of evaluation of food adulterants and toxic constituents of bakery products.
		PRACTICAL WORK:	
		1. Techniques of quality assessment of different natural and processed foods. 2. Documentation of details of baking ingredients, process and finished products used in baking industry. 3. Quality evaluation of milk and milk products 4. Sensory methods for measuring food quality assessment of raw materials used and processed bakery products 5. Instrumental methods for measuring food quality assessment of raw materials used and processed bakery products	

		6. Study of cleaning and sanitizers used in pre and post-operative processes in bakery industry 7. Listing of quality control agencies at national and international levels
12	**Suggested Resources for Reading**	1. InteazAlli (2003). Food Quality Assurance Principles and Practices.CRC Press. 2. J Andres Vasconcellos (2003) Quality Assurance for the Food Industry. A Practical Approach.CRC Press.
NOTE LATEST EDITION OF TEXT BOOKS MAY BE USED		

SEMESTER-II

<table>
<tr><td>1</td><td>Course Title</td><td>Food Packaging</td></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM-201</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration / LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 6 0 hours of practical.
(1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) (1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
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</table>
100 Marks (80 TH / PR + 20 IA)
a) The exam of this course will be of 3 hours duration.
b) There will be no objective type questions.
c) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td></td><td>Job Role</td><td>Dairy& Food Processor</td></tr>
<tr><td>7</td><td>Course Objective</td><td>1. Monitors packaging of various foods and their subsequent Compatibility and handles all categories of packaging such as primary, secondary and tertiary packaging for food products.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1 | To enable the students to understand about packaging and packaging materials, compatibility of various food items with packaging materials</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

11	**Course Content**	UNIT-I	Packaging Technology: Definitions, Functions of packaging, Properties of packaging material in relation to their functions, package design,Tests on packaging materials, Gas and water vapor transmission rates, types of containers- primary& secondary, flexible & rigid, hermetic & non-hermetic.
		UNIT-II	Packaging materials: Flexible films and pouches, Plastics used and their Specific applications. Rigid and Semi rigid plastic packaging methods. Retort pouch packaging. Laminated Paper board Cartons, Fiber Board and Corrugated Card Board packaging and their applications. Wood: structure, types of wooden containers and their properties. Glass : composition, properties, types and manufacture of glass containers.
		UNIT-III	Metal Cans: properties of metals for packaging material, Types of Metallic Cans - Tin cans and Aluminum cans, steel plate and functions of its various constituents, Importance of Open top sanitary cans, Lacqueringtypes and applications, three piece cans and two piece cans. Introduction to Canning operations – Can Reformer, Flanger, Seaming, Can closures, Sterilization of jars and bottles.
		UNIT-IV	Filling And Sealing Operations For Various Types of Packages: Closing and sealing of rigid plastic containers. Filling and sealing of Flexible plastic containers, Aseptic packaging, shrink packaging, gas packaging, vacuum and modified atmosphere packaging – principles, applications. Estimation of shelf life, analysis of storage requirement, accelerated storage Suggested Resources for Reading:
		PRACTICAL WORK:	
		1. To determine thickness of paper and paper board. 2. To identify different types of packaging materials. 3. Demonstration. of measurement of cartons' dimensions as per organizational standards. 4. Demonstration of measurement of dimensions of bottle mouth, cans and their caps. 5. Demonstration of sealing processes used in food industry. 6. Demonstration of filling process.	

12	**Suggested Resources for Reading**	1. Coles, R., Dowell, D.M., Kirwan, J. "Food Packaging Technology".Black Well Publishing 2. Ltd.ISBN084939788X, 2003. 3. NIIR Board: Food Packaging Technology Handbook National Institute of Industrial Research, New Delhi (2004) 4. Pirenger O.G.and A.L.Baver: Plastic Packaging Materials for Food Wiley VCH, GmbH, Germany (2000).
NOTE LATEST EDITION OF TEXT BOOKS MAY BE USED		

<table>
<tr><td>1</td><td>Course Title</td><td>Principles of food processing and preservation</td></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM-202</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration / LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 6 0 hours of practical.
(1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) (1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
100 Marks (80 TH / PR + 20 IA)
a) The exam of this course will be of 3 hours duration.
b) There will be no objective type questions.
c) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
</td></tr>
<tr><td></td><td>Job Role</td><td>Dairy & Food Processor</td></tr>
<tr><td>7</td><td>Course Objective</td><td>1. Working on different methods of processing of different foods and their preservation techniques.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
<table>
<tr><td>CO1</td><td>To understand the basic principles & objectives of food processing.</td></tr>
<tr><td>CO2</td><td>To study different means of food preservation and their subsequent utilization.</td></tr>
</table>
</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

11	**Course Content**	UNIT-I	Introduction- Historical developments of food preservation, Principles of food preservation, benefits of food preservation. Food Spoilage and to study its causes (microbial, physical or chemical)
		UNIT-II	Preservation by heat: Heat resistance of microorganisms, thermal death curve, heat treatments and their effects on food- Boiling, steam under pressure, blanching, pasteurization, canning, aseptic packaging, cans and container types, spoilage of canned foods, heat penetration in cans.
		UNIT-III	Preservation by low temperature- refrigeration storage, requirements of refrigerated storage, changes of foods during refrigerated storage, refrigeration load, Freezing and frozen storage- Slow and quick freezing, factors determining freezing rate, freezing methods, changes in foods during freezing, frozen food storage, Freezing curve, Intermediate moisture foods- their advantages and problems. Drying, dehydration and concentration- Types of drying, types of dryers, food concentration and their methods, changes in foods during dehydration and concentration.
		UNIT-IV	Microwave Heating- Properties, mechanism and its food applications. Preservation by radiationsUltraviolet and ionizing radiations, its applications in foods. Chemical Preservation: Types, uses and effects of class I & class II preservatives in foods.
		PRACTICAL WORK:	
		1. Study of blanching process in vegetables. 2. Dehydration of foods. 3. Preservation of food products by low temperature. 4. Use of chemicals in preservation of food. 5. Microwave cooking of food. 6. Cut out examination of canned foods. 7. Visit to food industry.	
12	**Suggested Resources for Reading**	1. Food Science by Potter NN, 2006, CBS Publishers, New Delhi. 2. Food Microbiology by Frazier WC & West Hoff DC, 1998, Tata Macgraw Hill Publishers, New Delhi. 3. 3. The Technology of food preservation by Desrosier and Desrosier.	

		4. Fellows, P. (2000) Food Processing Technology, Principles & Practices 2nd Edition Woodhead, Cambridge.
NOTE LATEST EDITION OF TEXT BOOKS MAY BE USED		

<table>
<tr><td>1</td><td>Course Title</td><td>Food Microbiology</td></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM-203</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration / LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 6 0 hours of practical.
(1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) (1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
100 Marks (80 TH / PR + 20 IA)
a) The exam of this course will be of 3 hours duration.
b) There will be no objective type questions.
c) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td></td><td>Job Role</td><td>Dairy & Food Processor</td></tr>
<tr><td>7</td><td>Course Objective</td><td>1. To identify various microorganisms associated with food and to analysis the microbiological quality of different food products.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
<table>
<tr><td>CO1</td><td>To understand the isolation methodology of microorganisms</td></tr>
<tr><td>CO2</td><td>To identify the microorganisms of food products of plant and animal origin.</td></tr>
<tr><td>CO3</td><td>To learn about Food borne diseases and microorganisms.</td></tr>
</table>
</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole</td></tr>
</table>

		syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.	
11	**Course Content**	UNIT-I	Introduction : Discovery of microbial world, application of microbiology and its relevance to food technology, theory of spontaneous generation, germ theory of disease, kosh's postulates, pure culture concept, nature and properties of prokaryotic and eukaryotic.
		UNIT-II	Microscopy- light microscope- resolving power, magnification, bright field, dark field, electron microscopy, scanning electron microscope
		UNIT-III	Food as nutrients for various micro-organism, factors affecting the growth and survival of microorganisms in food., general features and importance of different group of bacteria, yeast and molds important in food, methods of microbial examination of food and food products. Definition of growth, growth cycle, growth rate, generation time, measurement of growth.
		UNIT-IV	Food spoilage- microbial and biochemical aspects of food spoilage, role of bacteria, yeast and molds in food spoilage, spoilage of cereals and cereal products, fruits and vegetables, meat and meat products, milk and milk products, food borne illness, food intoxication and food infection, bacterial food poisoning by streptococcus aureus, clostridium botulinum, salmonella, E. Coli, Clostridium perfringes, Listeria monocytogenes, Campylobacter jejuni
		PRACTICAL WORK:	
		1. To study different parts of microscope. 2. Study the instruments (autoclave, hot air oven, incubation, laminar flow, pH meter, and spectrophotometer) of microbiological laboratory. 3. Sterilization and disinfection of equipments used in food microbiology laboratory. 4. To determine the no. of microorganism with haemocytometer. 5. Isolation of fungi from food material. 6. Preparation of media, slant and broths required in microbial analysis of food. 7. Study of incubation tests of heated canned food.	

		8. Micro biological analysis of egg, cereals products and food products. 9. Determination of TPC in food products. 10. Determination of Coliform counts. 11. Microbiological analysis of water-samples.
12	**Suggested Resources for Reading**	1. Microbiology by Pelczar M.J., Chan E.C.S & Krieg N.R., McGraw Hill Co. 2. General Microbiology by Stanier R.Y., In graham J.L., Wheelies M.L. & Painter P.R. Macmillan press Ltd., London 3. Laboratory Manual in Microbiology by GunaSeKaran P, New age international Ltd., New Delhi. 4. Frazier WC and Westoff DC "Food Microbiology" Tata macro Hill Publishing 5. Adams MR and Moss MO "Food Microbiology"New age international Ltd,
NOTE LATEST EDITION OF TEXT BOOKS MAY BE USED		

ESSENTIAL COMPONENTS OF COURSE OF
B.VOC. (FOOD PROCESSING & QUALITY MANAGEMENT)
(SEMESTER III & IV)

PROGRAMME OUTCOMES

PO1. To enhance the capability of students to identify, analyse and solve to problems arising in food industries in the process of preparation and preservation of foods..

PO2. To strengthen the foundation of students to built up their career in food industry or to persue career in food as well as interdisplinary areas or to establish their entrepreneurship ventures

PO3. To equip students with knowledge to get job in government sector and private field.

PO4. To make students understand various handling, storage, processing, packaging and entrepreneurship techniques along with the environmental challenges.

SEMESTER-III

<table>
<tr><td>1</td><td>Course Title</td><td>Post-Harvest Management of Frutis & Vegetables</td></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM-301</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration / LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
100 Marks (80 TH / PR + 20 IA)
a) The exam of this course will be of 3 hours duration.
b) There will be no objective type questions.
c) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td></td><td>Job Role</td><td>Fruit & Vegetable Manufacturer</td></tr>
<tr><td>7</td><td>Course Objective</td><td>1. Students shall know about structure and composition of fruits and vegetables, various post-harvest losses encountered, their prevention, harvesting methods, importance of maturation and its determination. Students also study about ripening and shelf life of fruits and vegetables.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
<table>
<tr><td>CO1</td><td>Familiarize themselves with different aspects of post-harvest technology and recent developments</td></tr>
<tr><td>CO2</td><td>Understand the correlation of physiology and biochemistry of fresh produce with storage issues..</td></tr>
<tr><td>CO3</td><td>Measure respiration rate of fresh produce quantatively.</td></tr>
<tr><td>CO4</td><td>Know about extension shelf life for fresh produce.</td></tr>
</table>
</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

10	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.	
11	**Course Content**	UNIT-I	Definition and importance of Post-harvest technology in India. Chemical composition, Classification and nutritive value of fruits and vegetables. Post-harvest losses of fruits and vegetables and factors affecting them. Factors affecting post-harvest losses. Prevention of post- harvest losses.
		UNIT-II	Definition of harvesting, importance of harvesting. Harvesting methods- Manual and mechanical. Maturity indices of fruits and vegetables, importance of maturity indices, determination of maturity indices. Physiology of growth and development of fruits, climacteric and non-climacteric fruits.
		UNIT-III	Fruit ripening, Changes during ripening, Ethylene biosynthesis, Artificial ripening. Cleaning, Sorting and Grading of fruits and vegetables. Post-harvest diseases of fruits and vegetables.
		UNIT-IV	Treatment to enhance shelf life of fruits and vegetables. Physical- precooling, waxing, curing, packaging, Irradiation, Thermal treatment. Chemical treatment, fungicides and growth regulator. Transportation methods .Refrigerated and non-refrigerated. Traditional methods of storage- Pit storage, high altitude cooling, cellar and underground, Zero energy cool chamber (ZECC).
		PRACTICAL WORK:	
		1. Analyze the maturity stages of fruits & vegetables. 2. Dehydration & sun-drying of fruits & vegetables. 3. To study the ripening of fruits & vegetables.	

		4. To study the shelf-life of fruits & vegetables at low temperature. 5. To study the different types of spoilage in fruits & vegetables. 6. To study the effect of wax-coating on shelf-life of fruits & vegetables.
12	**Suggested Resources for Reading**	1. Food Facts & Principles by Shakuntla Manay N & Shadoksharaswamy N, 1996, New Age world publisher, CA. 2. Food Science by Potter, N.N.,CBS Publisher, New Delhi. 3. Preservations of fruits and vegetables by Girdhari Lal, Sidappa GS & Tandon GL, 1960, ICAR, New Delhi.
NOTE LATEST EDITION OF TEXT BOOKS MAY BE USED		

<table>
<tr><td>1</td><td>Course Title</td><td>Egg. Poultry Fish Processing</td></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM-302</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration / LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory +6 0 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
100 Marks (80 TH / PR + 20 IA)
a) The exam of this course will be of 3 hours duration.
b) There will be no objective type questions.
c) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td></td><td>Job Role</td><td>Egg. Poultry Fish Technologist</td></tr>
<tr><td>7</td><td>Course Objective</td><td>1. To enable the students to understand the various aspects of egg and fish products and their preparation.
2. To gain knowledge on processing of egg & fish products.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
<table>
<tr><td>CO1</td><td>Know about the commercial aspects of egg processing.</td></tr>
<tr><td>CO2</td><td>Learn about types and breeds of poultry important for industrial production, anti and post-mortem changes .</td></tr>
<tr><td>CO3</td><td>Understand waste utilization and disposal of poultry by-products.</td></tr>
<tr><td>CO4</td><td>Know about commercial processing of fish & fish processing products.</td></tr>
</table>
</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including</td></tr>
</table>

		Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.	
11	**Course Content**	UNIT-I	Egg-structure composition, nutritive value, interior quality of egg & its evaluation, functional properties, grading, microbial spoilage of egg & preservation and storage of eggs. Handling & transportation of egg.
		UNIT-II	Poultry types- Hen, turkey, ducks, geese. Anti-mortem examination & slaughtering of hen, poultry cut-up parts, poultry sanitation & waste disposal, MAP of poultry.
		UNIT-III	Fish- types of fish, composition, structure and nutritive value, postmortem changes in fish, on board handling, storage & transportation of fish.
		UNIT-IV	Curing, Smoking, salting, canning, freezing & drying of fish. Comminuted fish products, fish protein concentrates, fish-oil & fish meal. Packaging of fish.
		PRACTICAL WORK:	
		1. Determination of egg components. 2. Grading & quality evaluation of egg. 3. Preparation of egg products- boiled, scrambled & omelet. 4. Preservation of shell eggs. 5. Determination of egg density. 6. Visit to poultry farm.	
12	**Suggested Resources for Reading**	1. G.J. Mountney.1995. Poultry Products Technology by Taylor & Francis, 2. Parkhurst& Mountney.2012. Poultry Meat and Egg Production Springer London, Limited, 05-Jul2012 3. Food Facts & Principles by Shakuntla Manay N & Shadoksharaswamy N, 1996, New Age world publisher, CA. 4. Egg Science & Technology by Stadelman WJ, & Cotterill OJ, 2002, CBS Publisher, New Delhi. 5. Fish Processing & preservations by Charles L, Cutting.	
NOTE LATEST EDITION OF TEXT BOOKS MAY BE USED			

<table>
<tr><td>1</td><td>Course Title</td><td>Cereal, Milling & Legumes</td></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM- 303</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration / LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory +6 0 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
100 Marks (80 TH / PR + 20 IA)
a) The exam of this course will be of 3 hours duration.
b) There will be no objective type questions.
c) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td></td><td>Job Role</td><td>Cereal & Legumes Specialist</td></tr>
<tr><td>7</td><td>Course Objective</td><td>1. Students will understand different types of cereals, their structure, milling methods, quality evaluation, composition and milling methods of pulses and anti-nutritional factors present in them and their elimination.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
<table>
<tr><td>CO1</td><td>Understand structure & composition of major cereals.</td></tr>
<tr><td>CO2</td><td>Get knowledge about milling of different grains.</td></tr>
<tr><td>CO3</td><td>Learn about processing of pulses.</td></tr>
<tr><td>CO4</td><td>CO4. Assess quality of different cereal grains.</td></tr>
</table>
</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.</td></tr>
</table>

		3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.	
11	**Course Content**	UNIT-I	Introduction to cereal grain & different types of grains, Milling criteria & quality criteria for grains, Wheat- structure, chemical composition, types & milling. Wheat cleaning & conditioning, Traditional & modern milling of wheat and flour extraction rate.
		UNIT-II	Flour grades, treatment & storage. Rice- structure & chemical composition, traditional & modern- milling of paddy.
		UNIT-III	Structure & chemical composition of maize, Wet & dry milling of corn. Barley- types, milling, malting & brewing of barley. Oat- structure & milling
		UNIT-IV	Introduction and chemical composition of pulses, Milling and de-cortication & polishing of pulses, anti-nutritional factors present in pulses and their elimination.
		PRACTICAL WORK:	
		1. Determination of gluten. 2. Determination of physical characteristics of wheat. 3. Determination of moisture, ash in cereal grains. 4. Determination of physical characteristics of Rice. 5. Milling of wheat. 6. Cooking quality of rice. 7. Visit to Cereal industry.	
12	**Suggested Resources for Reading**	1. Technology of Cereals by Kent N. L. and Evers AD, 4th Ed., 1983, Woodhead Publishing Ltd., UK. 2. Principle of Cereal Science & Technology by Kent. NL, 1983, Pergamon Press, London, UK. 3. Food Facts & Principles by Shakuntla Manay N & Shadoksharaswamy N, 1996, New Age world publisher, CA. 4. Food Science by Potter, N.N.,CBS Publisher, New Delhi.	
NOTE LATEST EDITION OF TEXT BOOKS MAY BE USED			

SEMESTER-IV

<table>
<tr><td>1</td><td>Course Title</td><td>Technology of Oils & Fats</td></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM-401</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration / LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
100 Marks (80 TH / PR + 20 IA)
a) The exam of this course will be of 3 hours duration.
b) There will be no objective type questions.
c) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td></td><td>Job Role</td><td>Oils & Fats Processor</td></tr>
<tr><td>7</td><td>Course Objective</td><td>1. To understand the basic properties of oils & fats.
2. To know their nutritional importance and extraction of oils.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
<table>
<tr><td>CO1</td><td>Understand nomenclature and physiochemical properties of oils & fats.</td></tr>
<tr><td>CO2</td><td>Know about nutrition involved in oils & fats.</td></tr>
<tr><td>CO3</td><td>Learn about extraction methods in oils & fats.</td></tr>
<tr><td>CO4</td><td>Get aware about various storage issues of oils & fats.</td></tr>
</table>
</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole</td></tr>
</table>

		syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.	
11	**Course Content**	UNIT-I	Introduction to oils and fats, physical and chemical properties of oils and fats, Nutritional importance of oils and fats.
		UNIT-II	Extraction of oils/fats, refining, degumming, bleaching and deodourization, problems during storage (rancidity and reversion).
		UNIT-III	Hydrogenation of oils, fractionation & winterization of oils, functions of oils and fats in food processing: Quality assessment tests of fats and oils.
		UNIT-IV	Sources and physic-chemical properties of following oils: animal (butter oil, lard and Tallow) and plant(groundnut, sunflower, soyabean and coconut oil)
		PRACTICAL WORK:	
		1. To determine moisture content of oilseeds. 2. To determine FFA (free Fatty acid) of oil. 3. Detection of adulteration of fats or oil. 4. Visit to vegetable oil factory. 5. Determination of melting point of fats.	
12	**Suggested Resources for Reading**	1. Food Chemistry by Meyer LH,2006, CBS Publisher, New Delhi. 2. Food Science by Potter NN,5th Ed, 2006, CBS publisher, New Delhi. 3. Food Oils & Fats: Technology, Utilization and Nutrition by Lawson H, 1995, CBS Publisher, New Delhi.	
NOTE LATEST EDITION OF TEXT BOOKS MAY BE USED			

<table>
<tr><td>1</td><td>Course Title</td><td>Fruits & Vegetables Processing</td></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM-402</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration / LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory +6 0 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
100 Marks (80 TH / PR + 20 IA)
a) The exam of this course will be of 3 hours duration.
b) There will be no objective type questions.
c) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td></td><td>Job Role</td><td>Fruits & Vegetables Processor</td></tr>
<tr><td>7</td><td>Course Objective</td><td>1. To know technical details of processing of different fruits & vegetables in accordance with their composition.
2. To understand the various aspects of fruits & vegetables preservation.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
<table>
<tr><td>CO1</td><td>Understand various steps involved in processing of different products.</td></tr>
<tr><td>CO2</td><td>Learn about standardization and formulation of different products.</td></tr>
<tr><td>CO3</td><td>Know about available commercial products.</td></tr>
<tr><td>CO4</td><td>Learn processing of tomatoes in the form of different products.</td></tr>
</table>
</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.</td></tr>
</table>

		2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.	
11	**Course Content**	UNIT-I	Classification, chemical composition & nutritional value of fruits & vegetables, Preparation of fruits & vegetables for processing-washing, sorting, grading, peeling, blanching, grating, destining, pitting. Bottling & canning of fruits & vegetable products.
		UNIT-II	Packaging of whole fruits & vegetables, Processing and packaging of cut fruits & vegetables, general methods of freezing of fruits & vegetables, their packaging & storage. Drying of fruits & vegetables.
		UNIT-III	FPO Specifications and preparation of Jam, Jellies, marmalade, preserves, pickles. Tomato processing-FPO standards and preparation of tomato juice, puree, paste, chutney, sauce and ketchup. Preparation and standards of fruit juices, squashes & cordials, fruit syrups, nectar, RTS & pulp.
		UNIT-IV	Machinery for peeling, slicing/dicing, pulping, grating, hydraulic pressing & Clarification. Fruit juice aroma recovery (Introduction & importance). Minimal processing- basic concepts.
		PRACTICAL WORK:	
		1. Preparation of pickles (sweat & sour). 2. Preparation of fruit juices. 3. Preparation of tomato ketchup & its preservation. 4. Preparation of Amla preserve & candy. 5. Minimal processing of fruits and vegetables. 6. Organoleptic evaluation of fruit & vegetable products. 7. Visit to fruit & vegetable industry.	
12	**Suggested Resources for Reading**	1. Preservations of fruits and vegetables by Girdhari Lal, Sidappa GS & Tandon GL, 1960, ICAR, New Delhi. 2. Food Facts & Principles by Shakuntla Manay N & Shadoksharaswamy N, 1996, New Age world publisher, CA. 3. Food Science by Potter, N.N.,CBS Publisher, New Delhi. 4. Shrivatsava, R. P. & Kumar S, 1998. "Fruit and vegetable preservations. Principles & practices".	

		International Book Distributing Co. Company. Lucknow.
NOTE LATEST EDITION OF TEXT BOOKS MAY BE USED		

<table>
<tr><td>1</td><td>Course Title</td><td>Processing & Meat & Meat products</td></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM-403</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration / LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory +6 0 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
100 Marks (80 TH / PR + 20 IA)
a) The exam of this course will be of 3 hours duration.
b) There will be no objective type questions.
c) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
</td></tr>
<tr><td></td><td>Job Role</td><td>Meat Specialist</td></tr>
<tr><td>7</td><td>Course Objective</td><td>1. Students will know about scope of meat, processing industry, composition and structure of meat, classification, meat quality, ante-and Post mortem change in meat, various techniques involved in meat processing.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
<table>
<tr><td>CO1</td><td>Understand the structural composition, types and quality of meat.</td></tr>
<tr><td>CO2</td><td>Know about various techniques of slaughtering animals.</td></tr>
<tr><td>CO3</td><td>Learn about different preservation techniques of meat.</td></tr>
<tr><td>CO4</td><td>Know novel products available in meat industry.</td></tr>
<tr><td></td><td></td></tr>
</table>
</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.</td></tr>
</table>

		2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.	
11	**Course Content**	UNIT-I	Meat- scope of meat processing industry in India, structure, composition & nutritive value of meat, classification of meat- mutton, pork & sheep. Meat quality parameters- meat colour, water holding capacity, marbling, firmness and factors affecting it.
		UNIT-II	Slaughtering and dressing of meat animal, anti-mortem examination of animal meat post mortem changes in meat- rigor mortis, biochemical changes associated with rigor mortis, conversion of muscle to meat.
		UNIT-III	Processing of meat-aging of meat, meat tenderization, (natural and artificial methods), methods of meat tenderization, factors affecting tenderness, storage of meat, microbial spoilage of meat, meat by-products.
		UNIT-IV	Preservation of meat- Chilling, Freezing, Canning, Curing, drying & smoking. Cooking, palatability and eating quality of meat, restructured meat products.
		PRACTICAL WORK:	
		1. Study of post-mortem changes. 2. Preparation of sausages, meat-balls and kabab. 3. Pre-slaughter of operation of meat animals. 4. Visit to slaughter-house. 5. Evaluation of meat quality. 6. Slaughtering and dressing of meat animals. 7. Preservation of meat by different methods.	
12	**Suggested Resources for Reading**	1. R.A. Lawrie, 1988 Meat Science, Pergamon Press. 2. Food Facts & Principles by Shakuntla Manay N & Shadoksharaswamy N, 1996, New Age world publisher, CA. 3. Food Science by Potter, N.N.,CBS Publisher, New Delhi.	
NOTE LATEST EDITION OF TEXT BOOKS MAY BE USED			

<table>
<tr><td>1</td><td>Course Title</td><td>Processing of Cereals</td></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM-404</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration / LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory +6 0 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
100 Marks (80 TH / PR + 20 IA)
a) The exam of this course will be of 3 hours duration.
b) There will be no objective type questions.
c) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td></td><td>Job Role</td><td>Cereals Processor</td></tr>
<tr><td>7</td><td>Course Objective</td><td>1. Students will know about quality evaluation of cereals by using various instruments, bakery ingredients and their role in baking, preparation of various bakery products, parboiling of rice. Processing of rice bran oil and various other cereal products.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
<table>
<tr><td>CO1</td><td>Evaluate quality of wheat flour.</td></tr>
<tr><td>CO2</td><td>Learn manufacturing process of different bakery products, corn products and rice brain oil.</td></tr>
<tr><td>CO3</td><td>Know about parboiling of rice and its benefits.</td></tr>
<tr><td>CO4</td><td>Learn about millet products and their processing.</td></tr>
</table>
</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type</td></tr>
</table>

		covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.	
11	**Course Content**	UNIT-I	Wheat flour- types and usage, improver and bleachers- their principle and actions, raising and thickening agents- their uses. Leavening agents- their role in baking industry. Quality criteria for wheat flour.
		UNIT-II	Physical dough testing instruments, major and minor ingredients used for bakery products. Preparation methods of bread, cookies and cakes.
		UNIT-III	Parboiling of paddy**,** methods, advantages and disadvantages and various changes during parboiling. Extraction of rice bran oil and its use.
		UNIT-IV	Manufacture of corn flakes, tortilla and corn syrup from corn starch. Preparation of oat meals and oat grits. Preparation of beer and ready to eat millet products.
		PRACTICAL WORK:	
		1. Preparation of White Bread. 2. To determine moisture in bread and biscuits. 3. Preparation of oat products. 4. Preparation of millet cookies. 5. Principle and preparation of Fruit cake. 6. Determination of dry and wet gluten in flour.	
12	**Suggested Resources for Reading**	1. Technology of Cereals by Kent N. L. and Evers AD, 4th Ed., 1983, Woodhead Publishing Ltd., UK. 2. Principle of Cereal Science & Technology by Kent. NL, 1983, Pergamon Press, London, UK. 3. Bread: A baker's book of techniques and recipes by Jeffrey Hamelman 4. Bakery-I, Student handbook and practical manual, published by CBSE. 5. A professional Text to bakery and confectionary by John Kingslee 6. Food Facts & Principles by Shakuntla Manay N & Shadoksharaswamy N, 1996, New Age world publisher, CA. **7.** Food Science by Potter, N.N.,CBS Publisher, New Delhi.	
NOTE LATEST EDITION OF TEXT BOOKS MAY BE USED			

Semester V

<table>
<tr><th>1</th><th>Course Title</th><th>Technology of fermented foods -I</th></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM- 501</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
100 Marks (80 TH/PR + 20 IA)
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Fermentation specialist</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>Study of various fermented products and role of different microorganisms in bringing about fermentation in foods.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. To understand the process of fermentation in foods
CO2. To study the different types of fermentations.
CO3.To study different types of microorganisms used to bring about fermentation in different foods.
CO4. To know variety of foods that can be obtained by the process of fermentation
CO5. To understand the therapeutic value of fermented foods.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

12 Course Content:

UNIT –I

Introduction to fermentation, concept of fermented foods, scope and development of fermented foods, rate of microbial growth and death, benefits of fermented foods**.**

UNIT –II

General methods of fermentation- Aerobic fermentation, Anaerobic fermentation. Classification of food fermentation- alcoholic, acetic & lactic fermentation.
Types of fermentation system (submerged surface and solid state)

UNIT III

Component parts of a fermentor and their functions, contamination problems in fermentor**.**
Food fermentations- Lactic acid fermentation of milk, vegetables, cereals and meat.

UNIT IV

Fermented milk products- curd, Yoghurt, acidophilus milk, bulgaricus milk, kefir and kumiss.

Practical Work:

1. Visit to food industries.
2. To study different parts and working of fermentor.
3. Barley steeping, germination, malting, mashing and brewing of malt
4. Preparation of sauerkraut.
5. Preparation of idli and dosa
6. Preparation of fermented milk products**.**

13 Suggested Resources for Reading

1.Industrial-Microbiology by Prescott & Dunn
2. Food facts and principles by Shakuntala Manay N &Shadoksharaswamy N, New age world publishers.
3. Principles of fermentation technology by Stanburry and Whittaker
4Potter NN Food Science CBS Publishers ND
5. Frazier WC and Westoff DC 'Food Microbiology" Tata Macgrow Hill publishing**.**

Note: Latest edition of text books may be used.

<table>
<tr><th>1</th><th>Course Title</th><th>Food analysis</th></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM -502</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
100 Marks (80 TH/PR + 20 IA)
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food analyst</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To study the analysis of food by measuring different parameters of foods.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. To know various sampling techniques.
CO2. To study different techniques used in food analysis in food laboratories.
CO3. To learn to work on various instruments.
CO4. To study the calliberation and standardization of different equipments.
CO5. To study the principle and working of highly sophisticated instruments which give results in seconds.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

12 Course Content:

UNIT-I

Introduction to food analysis, types of samples and sampling techniques, collection, storage and preservation of samples, preparation of food sample.

UNIT-II

Sampling of cereals, fruits and vegetables, milk, eggs, meat, fish, dried, frozen food products and canned food products**.**

UNIT-III

Physico-chemical methods for food analysis moisture, fat, protein, carbohydrate, crude fibre, ash and its types, minerals and vitamins in food**.**

UNIT-IV

Calliberation and standardization of different instruments. Technology used in food analysis; principle, types and applications of calorimetry, spectroscopy, photometry, chromatography and electrophoresis, refractometry, polarimetry and atomic absorption spectrophotometry **(AAS).**

Practical work:

1. Determination of moisture content in food samples
2. Determination of ash content in food samples
3. Determination of fat content in food samples.
4. Demonstration of instruments- farinograph, UV-VIS spectrophotometer, microscope.
5. Determination of moisture content using instant moisturometer.
6. Visit to food laboratory.

13 Suggested Resources for Reading:

1 Ranganna. S., Handbook of Analysis & Quality Control for fruits & vegetable Products, Tata Mcgraw Hill, New Delhi.

2. Jacobs.M.B, Chemical Analysis of Food & Food products, CBs Publishers, New Delhi.

Note: Latest edition of text books may be used.

<table>
<tr><td>1</td><td>Course Title</td><td>Neutraceuticals & Functional foods</td></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM-503</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
100 Marks (80 TH/PR + 20 IA)
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Quality analyst</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>Study of health benefits beyond basic nutrition, focusing on disease prevention, management and health promotion.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Understand health benefits of various bioactive compounds
CO2. Understand disease prevention for heart, brain and gut.
CO3.Study of food processing , safety and regulatory compliance.
CO4. Formulation of market foods that provide health beyond basic nutrition.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interacti.ve sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

12 Course Content:

UNIT-I

Definition, neutraceuticals compounds and their classification based on chemical/biochemical nature with suitable description.
Scope and future prospects, Relation of neutraceutical science with food technology

UNIT-II

Role of Neutraceuticals in health and disease prevention. Dietary fibers and prebiotics, Probiotics and symbiotics**.**

UNIT -III

Neutraceuticals Vs functional foods Vs dietary supplements
Definition of functional foods and their description. Sources of functional foods (plants, animals and microbial)

UNIT-IV

Functional Foods- Yoghurt, Fortified milk, protein bars, whey based beverages,
Fruit Juices fortified with vitamins/minerals, herbal tea, fortified breakfast cereals

Practical Work:

1. Identification of neutraceutical and functional food available in market.
2. Visit to a departmental store and pharmacy.
3. Preparation of any functional food in laboratory.
4. Preparation of protein bar in laboratory.
5. Study of food law dealing with regulations of functional foods, neutraceticals and dietary supplements.

13 Suggested Resources for Reading:

1. Functional foods and neutraceuticals by Karaikudi.
2. Functional foods and Neutraceuticals by Rotimi E. Aluko (springer publisher**)**
3. Food Science by B Srilakshmi**.**

Note: Latest edition of text books may be used.

<table>
<tr><th>1</th><th>Course Title</th><th>Technology of spices and flavors</th></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM-504</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practical.
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
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<tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
100 Marks (80 TH/PR + 20 IA)
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Spice and flavor quality analyst</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To study the quality and purity of various spices and flavors to be used in food products.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Understanding various types of spices, flavors and their characteristics.
CO2. Study of techniques of post-harvest management and processing of major and minor spices.
CO3. Study of value-addition and product development of spices.
CO4. Study of quality standards, control and safety of spices
CO5. Study of applications of flavor technology.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

12 Course Content:.

UNIT I

Classification and uses of spices, chemical constituents of spices, microbial contamination and insect infestation in spices and its control

UNIT –II

Chemistry & Technology of major spices-pepper, cardamom, ginger and turmeric.

UNIT-III

Chemistry & technology of minor spices- cumin, coriander, cinnamom, fenugreek, garlic, cloves and vanilla.

UNIT-IV

Cryomilling of spices, spice oleoresins and spice emulsion, essential oils, packaging of spices and spice products. Processing of cocoa and coffee, classification of flavouring compounds**.**

Practical work:

1. Identification of different spices.
2. Determination of moisture in ground spices.
3. Determination of total ash in spices.
4. Determination of extraneous matter in spices.
5. Adulteration tests of different spices
6. Organoleptic evaluation of flavors.

13 Suggested Resources for Reading

1. Handbook on spices, National Institute of Industrial Research (NIIR) board, Asia Pacific Business Press Inc. New Delhi 2004.
2. Spices and condiments by Pruthi, J. S., 1976.
3. Handbook of Spices by Peter K. V. 2001, Woodhead Publishers, UK.
4. Food Facts and Principles bt Shakuntla Manay and M. Shadaksharaswamy.

Note: Latest edition of text books may be used.

<table>
<tr><th>Course title:</th><th colspan="2">Unit operations in Food Engineering</th></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM 505</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
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</table>
100 Marks (80 TH/PR + 20 IA)
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food engineer</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To analyse the working and output of various equipments used in food processing industry.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. To create awareness about the unit operations involved in food processing industry.
CO2. To explain the principles of methods used for preservation of food.
CO3. To study different types of equipments used in the food processing industry.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
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3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

12 Course Content:

UNIT –I

Units and measurements- brief introduction to dimensions, fundamental units and derived units, system of measurements-cgs, mks & SI units

UNIT-II

Heat exchangers- brief introduction to principle of heat transfer (conductive, convective & radiative).

Steam injection and steam infusion, tubular, scraped surface, plate heat, shell & tube heat exchangers.

UNIT-III

Driers- Principles and application of tray driers, rotary drier, drum drier, fluidized bed drier, spray drier, vacuum drier and freezer drier**.**

UNIT-IV

Evaporators-different types of evaporators, principles of single effect and multiple effect evaporator, steam economy, thermal and mechanical vapour recompression system**.**

Practical work:

1. Study of psychometric charts –use and applications.
2. Determination of moisture content on wet and dry basis.
3. Study of dehydration characteristics of different food materials.
4. To study the working principle of an evaporator.
5. Shelf life evaluation of food product.

13 Suggested Resources for Reading

1. Fundamentals of Food processing engineering by Romeo T Taledo, CBS Publications.
2. Introduction to food engineering by R Paul Singh and Dennis R Heldman – Academic press London.
3. Unit operations of agriculture processing by K M Sahay and K K Singh ,Vikas Publishers.
4. Unit operations of chemical engineering by Mccab and Smith, Mcgraw Hill, New Delhi.

Note: Latest edition of text books may be used.

<table>
<tr><th>1</th><th>Course Title</th><th>Food Safety and Food Laws</th></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM- 506</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory +60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
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(1 Semester = 15 weeks)
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</table>
100 Marks (80 TH/PR + 20 IA)
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Quality analyst and Food safety officer</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To ascertain the safety of food by assessing the quality of food and their conformation to food laws</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. To understand the importance of food safety.
CO2.To study implementation of food safety systems
CO3. To study national and international food standard and laws
CO4. To have a detailed knowledge of implementation of food sanitation in the food premises.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

12 Course Content;

UNIT I

Introduction to Food safety –definition, factors affecting food safety, importance of safe foods. Food-borne pathogenic microorganisms, food poisoning/food intoxication**.**

UNIT-II

Food Safety Management System- Pre-requisites of Food Hygiene- GHP, GLP, GAP, GMP, HACCP, ISO-22000, TQM**.**

UNIT-III

Hygiene & sanitation- importance of personal hygiene of food handlers- habits, clothes, illness. Education in food handling and services. Cleaning agents & disinfectants. Food sanitation- Principles and methods, control & inspection.

UNIT-IV

Food Safety laws and regulations- detailed study of FPO, PFA, AGMARK, CODEX Alimentarius, MFPO, Weight and Measures Act**.**

Practical work:

1. Preparation of HACCP chart for food industry.
2. Quality evaluation of oils and fats.
3. Quality evaluation of fruits and vegetables.
4. Quality evaluation of meat and poultry.
5. Quality evaluation of cereal industry.
6. Quality evaluation of dairy industry.

13 Suggested Resources for Reading

1.Adam MR and Moss MO Food Microbiology New Age international (p) Ltd. ND.

2. Potter NN Food Science CBS Publishers ND.

3. Food Safety by Ian C Shaw Wiley Blackwell.

4. Principles of Food Sanitation by Marriot, CBS Publishers, New Delhi**.**

Note: Latest edition of text books may be used.

Semester VI

<table>
<tr><th>1</th><th>Course Title</th><th>Food Plant Layout</th></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM 601</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
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(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
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<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
100 Marks (80 TH/PR + 20 IA)
e. The exam of this course will be of 3hours duration.
f. There will be no objective type questions.
g. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
h. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food plant layout officer</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>Managing and updating food plant layout time to time as per business policies and safety standards</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1 Introduction of the basic setup of a food processing industry.
CO2.To make them conversant with the machinery and equipments used in different types of food industry.
CO3. To study the different types of food plant layout.
CO4. To study the design of layout of different types of industries.
CO5. To study the application of HACCP concept.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>5. The syllabus of this paper has been divided into FOUR units.
6. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
7. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
8. All questions carry equal marks.</td></tr>
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12 Course Content:

UNIT –I

Industrial plant design concepts & general design considerations for location of food plants. Application of HACCP concepts and ISO requirements.

UNIT II

Importance of plant layout, selection of site, selection of building material, selection and planning of manufacturing process & service facilities**.**

UNIT III

Types of layouts, their importance. Basic understanding of equipment layout and ventiliation in food processing plants**.**

UNIT IV

Process flow charts for material movement & utility consumption in food plants. Miscellaneous aspects of plant layout and design like provisions for waste disposal**,** safety arrangements etc

Practical Work:

1. Preparation of layout and process diagram of potato chip manufacturing plant
2. Preparation of layout and process diagram of Jam/marmalade manufacturing plant.
3. Preparation of layout and process diagram of bread making plant.
4. Preparation of layout and process diagram of dairy plant
5. Industrial visit and report making.

13 Suggested Resources for Reading:

1. JM Apple, Plant layout and material Handling , John Willey & sons, (1977).
2. Manufacturing facilities Design and Material handling by Fred E. Meyers, and Mathew P.Stephens, 3rd edition, Pearson Prentice Hall, 2000.
3. James M Moore, Plant Layout and Design, McMillan & Co.,(1959)

Note: Latest edition of text books may be used.

<table>
<tr><th>1</th><th>Course Title</th><th>Food Additives</th></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM-602</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
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(Practical)</td><td>30 hours of lectures for Theory +60 hours of practicals
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(1 Semester = 15 weeks)
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100 Marks (80 TH/PR + 20 IA)
e. The exam of this course will be of 3hours duration.
f. There will be no objective type questions.
g. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
h. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food analyst</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To analyse and assess the presence of various additives in different forms in different food products.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. To give the knowledge of various additives used in food products and their functions.
CO2. To study the properties of various food additives , their hazards and limits prescribed under food safety regulations.
CO3. Understanding the importance of various types of substances added to food for their fortification and enrichment.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>5. The syllabus of this paper has been divided into FOUR units.
6. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
7. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
8. All questions carry equal marks.</td></tr>
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12 Course Content:

UNIT I

Definition, classification and functions, nutritional and non-nutritional food additives, uses and functions of acid, base, buffers, salts and chelating/sequestering agents.

UNIT II

Antioxidants, emulsifying and stabilizing agents, anticaking agent, humectants, thickeners, firming agents, Flour bleaching agents and bread improvers**.**

UNIT III

Antimicrobial agent, Class I & class II preservatives, Food colours (synthetic and natural), pigments, their importance and utilization, flavouring agents (synthetic & natural) and related substances**.**

UNIT IV

Flavour emulsions, essential oils and oleoresins, application of fibre, clarifying agents.
Low calorie and non-nutritive sweeteners**.**

Practical work:

1. Description of quality/quantity recommended as safe(GRAS) food additives.
2. Applications of additives and ingredients in cereal foods.
3. Determination of adulterants in oils and fats and spices.
4. Food labelling with respect to nutrition and health.
5. To study inter-national codes for food additives**.**

13 Suggested Resources for Reading

1. Food Facts and principles by Shakuntala Manay N & Shadoksharaswamy N, New age world.
2. Potter NN Food Science CBS Publishers ND
3. Food Chemistry O.R. Fennema

Note: Latest edition of text books may be used.

<table>
<tr><th>1</th><th>Course Title</th><th>Fundamentals of Food Nutrition</th></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM-603</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
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</table>
100 Marks (80 TH/PR + 20 IA)
e. The exam of this course will be of 3hours duration.
f. There will be no objective type questions.
g. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
h. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Nutrition specialist</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>Study of various aspects of food nutrition, balanced diet, malnutrition and diet requirements of various age groups.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Get basic understanding about the identification of food sources and 5 food groups.
CO2. Perform the calculation of BMI and BMR.
CO3. Plan the diet planning of normal adult obese, underweight and overweight people.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>5. The syllabus of this paper has been divided into FOUR units.
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7. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
8. All questions carry equal marks.</td></tr>
</table>

12 Course Content :

UNIT-1

Introduction, Definition of food nutrition, Balanced diet and Malnutrition –Undernutrition and Overnutrition. Functions of food, food types and groups, water balance and Energy balance. Energy value of carbohydrates, fats and proteins, Calorific value of foods.

UNIT II

BMI and BMR of an individual, Factors affecting BMR, Dietary fibre its functions and sources. Acid-base Balance, Functions of nutrients –carbohydrates, fats, proteins, vitamins, minerals Recommended Dietary allowances for infants, children, adults and pregnant women**.**

UNIT III

Diet survey and Diet groups
Meal planning Food guide, objectives in meal planning & Meal planning Process
Importance of Therapeutic Nutrition, Planning of diet for patients suffering from ulcer, diarrhea, diabetes and jaundice.

UNIT IV

Disorders of nutrition, Protein-Energy Malnutrition (PEM), marasmus & Kwashiorkor
Deficiency diseases of Vitamins-night blindness, beri-beri, pellagra, scurvy & rickets.

Practical Work:

1. Identification of food sources for various nutrients.
2. Identification of 5 food groups.
3. Calorific value of foods.
4. Calculation of BMI and BMR.
5. Diet plan of people around you (normal adult, obese and underweight) people.

13 Suggested Resources for Reading:

1. Fundamentals of Foods, Nutrition and Diet Therapy by S.R. Mudambi &M.V. Rajagopal, 2012 New Age International Pvt. Ltd. Publishers. New Delhi.
2. Handbook of Food and Nutrition: by M. Swami Nathan Vol. I, II.2012, Bangalore Printing and Publishing.
3. Food Facts and principles by Shakuntala Manay N & Shadoksharaswamy N, New age world.
4. Potter NN Food Science CBS Publishers ND

Note: Latest edition of text books may be used.

<table>
<tr><th>1</th><th>Course Title</th><th>Food Beverage Technology</th></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM--604</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
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(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
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100 Marks (80 TH/PR + 20 IA)
e. The exam of this course will be of 3hours duration.
f. There will be no objective type questions.
g. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
h. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Beverage specialist</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>It focus on developing technical, scientific and operational expertise to process, preserve and manage food products efficiently and safely.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Focus on developing skills in food processing, quality control, safety (HACCP), engineering, microbiology and management to create safe, quality food /drinks.
CO2. Work in quality assurance, production or R & D in food beverage industry.
CO3.Become food safety officer or technical brewer.
CO4. Develop new products and innovative processes.
CO5. Manage food business or start their own ventures.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
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7. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
8. All questions carry equal marks.</td></tr>
</table>

12 Course Content:

UNIT I

Importance of Beverages and status of beverage industry, Processing of beverages packaged drinking water, juice based beverages.

UNIT II

Synthetic, still, carbonated, low-calorie and dry beverages
Dairy based beverages, alcohol based and fruit beverages, tea, coffee, cocoa etc

UNIT III

FSSAI specifications for beverages, Ingredients, manufacturing and packaging processes and equipment for different beverages
Miscellaneous beverages coconut water, sugarcane juice and flavoured syrups.

UNIT IV

Water treatment and quality of process water.
Sweeteners, colourants, acidulants, clouding and clarifying and flavouring agent for beverages.
Carbonation, quality tests of beverages.

Practical work:

1. Preparation of RTS beverage.
2. Preparation of artificial lemon juice.
3. Preparation of beverage using artificial sweetener.
4. Specification for different fruit beverages and preparation of fruit squash.
5. Measurement of pH and acidity of beverage.

13 Suggested Resources for Reading:

1. Food Facts and principles by Shakuntala Manay N & Shadoksharaswamy N, New age world.
2. Potter NN Food Science CBS Publishers ND.
3. Fruit and vegetable preservation by R.P.Srivastva and Sanjeev Kumar.
4. Preservation of fruit and vegetable products by GirdhariLal, Siddappa, Tondon

Note: Latest edition of text books may be used.

<table>
<tr><td>1</td><td>Course Title</td><td>Confectionary and sugar technology</td></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM-605</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
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(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
100 Marks (80 TH/PR + 20 IA)
e. The exam of this course will be of 3hours duration.
f. There will be no objective type questions.
g. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
h. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Sugar technologist or confectionary and candy maker</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>Designed to provide both theoretical knowledge and practical skills in the production of sugar based sweets and the refining of sugar</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Focus on providing students with the technical, operational and analytical skills needed to manufacture sugar based products and manage sugar refining processes.
CO2. Sugar analysis
CO3. Candy making
CO4. Chocolate and panning</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>5. The syllabus of this paper has been divided into FOUR units.
6. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
7. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
8. All questions carry equal marks.</td></tr>
</table>

12 Course Content:

UNIT I

Composition and characteristics of cane juice, cane Juice extraction, manufacturing of sugar, Deterioration of sugars during storage and transportation and its prevention.

UNIT II

By products of sugar industry and its utilization, Mollases and Bagasse.
Properties of Dextrose, fructose, lactose, caramel, maltose, honey, invert sugar and its characteristics**.**

UNIT III

Definition and classification of confectionary. Hard & soft boiled sugar confectionary- fondant, fudge, caramel, toffee and butterscotch.

UNIT-IV

Sugar panning, hard-boiled candy, lozenges, colours used in confectionary.
Packaging requirements of confectionary. Chocolates its manufacturing and defects.

Practical work:

1. Preparation of candies and caramel .
2. Collection of various types of confectionary packages.
3. Determination of refractive index of sugar solutions of different consistencies.
4. Organoleptic testing of different confectionary products.
5. Preparation of toffees.
6. Visit to confectionary store.

13 Suggested Resources for Reading:

1. Chocolate, cocoa and confectionary: Science and Technology by Minife, 1997, AVI Publishing Co, New York.
2. Handbook of Cane Sugar Technology by Mathur RBL, 1986 Oxford and IBH Publishing Co,, New Delhi.
3. Food Facts and principles by Shakuntala Manay N & Shadoksharaswamy N, New age world.
4. Potter NN Food Science CBS Publishers ND.

Note: Latest edition of text books may be used.

<table>
<tr><td>1</td><td>Course Title</td><td>Grain storage</td></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM-606</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory +60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
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100 Marks (80 TH/PR + 20 IA)
e. The exam of this course will be of 3hours duration.
f. There will be no objective type questions.
g. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
h. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Grain storage specialist</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>Focuses on maintaining grain quantity and quality through scientific management, covering principles of sanitation, pest control, moisture or temperature monitoring and safe handling to minimize post-harvest losses and ensure food security.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Focus on equipping students with the ability to maintain the quality, quantity and safety of stored grains.
CO2. Designing and conducting experiments for drying and stored grains
CO3. Understanding storage systems.
CO4. Quality maintenance of grains
CO5.Spoilage management and pest and disease control of grains.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>5. The syllabus of this paper has been divided into FOUR units.
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7. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.</td></tr>
</table>

8. All questions carry equal marks.

12 Course Content:

UNIT –I

General problems of grain storage, sources and detection of infestation in stored food grains. Causes, types and deterioration in stored food grains and method to check them.

UNIT II

Traditional and modern methods of bag and bulk storage. Insect pest of stored grains. Chemical, non-chemical and integrated methods of controlling stored grains insect pests**.**

UNIT III

Toxic contamination in good grains, their ill effects.
Pesticidial contamination tolerance limits, residue and precaution of safe handling of pesticides.

UNIT IV

Cleaning, aeration and drying of stored products at farmers, commercial and government levels. Role of moisture in spoilage of stored grains. Categorization of food grains for storage. Principle of godown sanitation and hygiene.

Practical work:

1. To study various insect pests of grains.
2. To store the grains and check its shelf-life.
3. To study the effect of moisture on spoilage on grains.
4. To study the quality tests for grains.
5. To study the various pesticides used for grain storage.

13. Suggested Resources for Reading:

1. Introduction of insect by Metalf & Lukemann.
2. Pesticides and pollution by Mollan.

Note: Latest edition of text books may be used.

<table>
<tr><th>1</th><th>Course Title</th><th>Enzymes in Food industry</th></tr>
<tr><td>2</td><td>Course Code</td><td>FPQM-607</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
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(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
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</table>
100 Marks (80 TH/PR + 20 IA)
i. The exam of this course will be of 3hours duration.
j. There will be no objective type questions.
k. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
l. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food specialist(baking, dairy, meat and fruit and vegetable).</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>Focus on understanding enzyme basics (kinetics, classification), production and application to improve food quality, yield and create noval products</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1.Understanding enzyme functions, classification and applications for improved texture, flavor and shelf-life of food products.
CO2. Students learn to apply enzymes to enhance food quality, digestibility and processing efficiency.
CO3. Quality control, product development and process control</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>9. The syllabus of this paper has been divided into FOUR units.
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11. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.</td></tr>
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12. All questions carry equal marks.

12 Course Content:

UNIT I

Enzymes- definition, classification, properties and mechanism of enzyme action, enzyme kinetics, nomenclature of enzymes. General applications of enzymes in food industry.

UNIT II

Enzymes in dairy industry, natural enzymes in milk, lactose intolerance, role of enzymes in cheese making and whey processing**.**

UNIT III

Enzymes for production of corn syrup solids, High fructose corn syrup.
Fruit Juices- cell wall degrading enzymes for liquefaction, clarification, debittering and decolorization**.**

UNIT IV

Baking- enzymes used in bread making (amylases), anti-staling. Enzymes used in dough conditioning.
Meat processing - meat tenderization, enzymes used in egg processing**.**

Practical work:

1. Study of effect of papain enzyme in meat tenderization.
2. Study the adequacy of pasteurization of milk.
3. Study the clarification of juices using enzyme.
4. Effect of temperature on enzyme activity.
5. Study of lipase activity of lipids.

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13 Suggested Resources for Reading:

1. Food Facts and principles by Shakuntala Manay N & Shadoksharaswamy N, New age world.
2. Tucker GA and Woods LFJ.1991 Enzymes in food processing**.**

Note: Latest edition of text books may be used.